JACK J. BURLESON

AI / MACHINE LEARNING • DATA SCIENCE

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PROFESSIONAL SUMMARY

M.S. candidate in Psychology: Data Science in Human Behavior at UW–Madison (expected Dec 2026). Applied AI practitioner building full-lifecycle systems: multi-agent document-intelligence pipelines, transformer fine-tuning and calibration, LLM evaluation, computer vision, and behavioral prediction. Capstone with American Family Insurance, developed in biweekly working sessions with its Enterprise Data Science lead. Five years of high-volume operations experience in quality assurance and process improvement.

EDUCATION

University of Wisconsin–Madison *Madison, WI* Expected Dec 2026

M.S. Candidate — Psychology: Data Science in Human Behavior · **GPA 3.85**

- Coursework in machine learning, statistics, behavioral data analysis, predictive modeling, and experimental design.

University of Wisconsin–Parkside *Kenosha, WI* Sep 2020 – May 2025

B.S., Psychology · Concentration: Neuroscience · **GPA 3.93**

- Sociology minor; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Madison Recreation & Wellbeing — Building Services *Madison, WI* Jun 2026 – Present

- Maintain campus recreation and wellbeing centers to high standards of cleanliness, safety, and readiness.
- Respond to facility and equipment issues across multiple locations, keeping centers safe during peak use.

Avis Budget Group — Driver *Madison, WI* Jun 2026 – Present

- Transport, stage, and safety-check rental vehicles across a 500+ vehicle fleet for on-time availability.
- Coordinate vehicle movements between locations while upholding fleet safety standards.

UW–Parkside — Undergraduate Researcher, Cognitive Neuroscience Lab *Kenosha, WI*
Jun 2024 – Aug 2025

- Conducted cognitive neuroscience research under Dr. Melissa Gregg, designing and pilot-testing a study of truth, deception, and cognitive workload (code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems and analyzed neural time-series data for lab research on sleep quality and cognitive load.
- Presented at the 2025 Midwestern Psychological Association conference (Chicago) and at Research in the Rotunda.

Meijer Distribution — Logistics Coordinator *Pleasant Prairie, WI* Jul 2020 – Aug 2025

- Promoted from Retail Loader to Logistics Coordinator for reliable quality assurance and process documentation.
- Analyzed QA data for 64 daily deliveries across a Midwest regional network, keeping error rates below 1%.
- Partnered with warehouse and transportation teams to resolve quality problems and exceptions before shipment.

TECHNICAL SKILLS

Languages & Data	Python (pandas, NumPy, SciPy, scikit-learn, PyTorch, Hugging Face, PyArrow), R, SQL, MATLAB, JavaScript/TypeScript, Parquet, Hive
Machine Learning	Transformer fine-tuning, neural networks, Lasso/Ridge, ensembles (random forests, bagging, XGBoost), resampling, calibration, cross-validation, SHAP, computer vision
AI Engineering	LangGraph (LangChain), multi-agent pipelines, LLM evals & LLM-as-a-judge, prompt engineering, vLLM, Ollama, LM Studio, OpenRouter, ONNX + FastAPI on Modal
Tools & Cloud	Git, Docker, pytest, bash, gRPC, Spark/PySpark, HDFS, CHTC, Quarto, AWS, Azure, GCP

PROJECTS

Digital-Mailroom *DSHB Capstone · American Family Insurance (industry partner)*

- 7 Python packages covering entity extraction, classification, routing, and scalable legal-document workflows, addressing document-intelligence problems scoped by American Family Insurance's Enterprise Data Science division.
- Built one LangGraph pipeline that joins legal-document routing, prompt experiments, deterministic scoring, corpus analysis, and deployment in a reproducible Docker environment, sending each document to the cheapest step that can handle it.

Mailroom-ML *ML Training, Evaluation & Fine-Tuning*

- Fine-tuned a ModernBERT document classifier to 95.05% document-type accuracy (307 of 323) and 0.0152 window-level expected calibration error on a test split held out from training and calibration, with data lineage tracking, per-head temperature scaling, and a confidence gate that decides which documents can skip an LLM.
- Audited the training data, found a label leak (42.8% of fallback filenames named the subclass), removed it, and added a build-time guard. Trained in PyTorch with Hugging Face datasets on Modal GPUs, evaluated with SciPy, exported to ONNX for FastAPI serving, and kept 485 pytest tests in CI.

Mailroom-Corpus-EDA *Data Engineering & Dataset Audit*

- Assembled a 3,302-document, 5-class mailroom corpus from public sources (CUAD, MAUD, SEC EDGAR, Enron email, synthetic insurance claims) with stable document IDs, content hashes, per-label provenance, and tracked licenses.
- Built a 7-phase audit pipeline that reruns with one command (integrity, composition, 48 figures, 17 tables), backed by about 200 tests; all 13,753 contract clause annotations match the document text exactly.

RVL-CDIP-Classifier *Computer Vision & Prompt Engineering*

- Built a Python toolkit that classifies scanned pages into the 16 RVL-CDIP categories with OpenRouter vision models at about $0.0004 per image, plus preprocessing that converts TIFF/PDF pages to 300-DPI images and runs Tesseract OCR.
- Iterated through 29 registered prompt versions using confusion matrices, misclassification analysis, and cost projections; the v11.8 prompt scored 82.6% on a 1,120-image class-balanced evaluation slice.

Psychometric Digital Twin *Behavioral ML Research · UW–Madison*

- Predicted communication-apprehension (PRCA) scores for 241 analyzed survey respondents (252 matched) from demographic, employment, geographic, and transit features with Ridge, Elastic Net, k-NN, random forest, XGBoost, and MLP models, using cross-validation and SHAP feature importance.
- Tested the PRCA's psychometric structure with factor analysis, then built LLM "digital twin" personas with tiered embodiment prompts (Ollama, OpenRouter, vLLM) and benchmarked them against the ML baselines on shared metrics (MAE, band accuracy, F1, AUC); the personas did not beat the classical models, and their bias was analyzed by demographic group.

Langton's Ant *Simulation · TypeScript*

- Built a browser simulator for Langton's ant and other turmites: any rule up to 32 colors and 16 states on an unbounded, chunked grid, with a gallery of 20 published machines and shareable permalinks.
- Kept the engine deterministic (integer-only stepping, seeded randomness) and covered it with 99 passing tests, including a pinned determinism test.

RESEARCH TRAINING

CITI Program: Conflicts of Interest 2025

CITI Program: IRB Training for Students and Faculty 2025